

Client spec cards

Hand each team one client card. Every design must meet every row. A 1:12 running slope means 12 cm of **horizontal run** for every 1 cm of rise; the sloped deck is slightly longer. Cut the deck from the 76.2 cm side of one 20 × 30 in foam board. Make tape hinges for folding, then brace them with the listed sticks, dowels, or binder clips for the load test. Hang weights at mid-span; use the unwound toy car only for the separate roll check.

Client A Community school

Rise to cover	5 cm
Maximum running slope	1:12
Minimum horizontal run	60 cm
Deck cut size	61 × 15 cm
Mid-span load test	1 × 200 g = 200 g
Portability test	folds flat: 2 panels; max 31 cm each

Client B Public library

Rise to cover	4 cm
Maximum running slope	1:14
Minimum horizontal run	56 cm
Deck cut size	57 × 15 cm
Mid-span load test	2 × 200 g = 400 g
Portability test	folds flat: 2 panels; max 29 cm each

Client C Health clinic

Rise to cover	6 cm
Maximum running slope	1:12
Minimum horizontal run	72 cm
Deck cut size	73 × 15 cm
Mid-span load test	3 × 200 g = 600 g
Portability test	folds flat: 3 panels; max 25 cm each

Client D Science museum

Rise to cover	4 cm
Maximum running slope	1:16
Minimum horizontal run	64 cm
Deck cut size	65 × 15 cm
Mid-span load test	4 × 200 g = 800 g
Portability test	folds flat: 2 panels; max 33 cm each

Instructor math check:

Client A: run $5 \times 12 = 60$ cm; deck $\sqrt{(60^2 + 5^2)} = 60.2$ cm, cut 61 cm; folded panel $61 \div 2 = 30.5$ cm

Client B: run $4 \times 14 = 56$ cm; deck $\sqrt{(56^2 + 4^2)} = 56.1$ cm, cut 57 cm; folded panel $57 \div 2 = 28.5$ cm

Client C: run $6 \times 12 = 72$ cm; deck $\sqrt{(72^2 + 6^2)} = 72.2$ cm, cut 73 cm; folded panel $73 \div 3 = 24.3$ cm

Client D: run $4 \times 16 = 64$ cm; deck $\sqrt{(64^2 + 4^2)} = 64.1$ cm, cut 65 cm; folded panel $65 \div 2 = 32.5$ cm. All cut lengths are at most 76.2 cm; a 15 cm-wide deck leaves 35.8 cm of board width for supports.